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REGULAR PAPER

Interpretable and robust multimodal fake news detection with evidence fusion and reliable reasoning

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Abstract

Multi-modal fake news detection involves the complex task of jointly analyzing textual and visual information, whereas uni-modal systems often under-perform due to limited contextual understanding. This paper presents a reliable and interpretable multi-modal framework that integrates semantic and stylistic cues using BERT for textual encoding, and CLIP and SWIN Transformer models for visual representation. Intra-modal features are fused via a Dempster–Shafer theory-based gating mechanism, enabling uncertainty-aware reasoning within each modality. Cross-modal features are aligned and refined using transformer-based correlation modeling to capture nuanced interactions between modalities. The framework is further strengthened by a composite loss function, formed by combining evidential deep learning with supervised and unsupervised contrastive learning strategies. This results in improving the classification accuracy and enhancing the model's robustness by quantifying uncertainty and allowing the rejection of low-confidence predictions. The inclusion of the uncertainty regularization ensures well-calibrated outputs, which are crucial for real-world deployment where interpret-ability and trust are essential. Extensive experiments on the Fakeddit dataset demonstrate that the proposed approach consistently outperforms strong baselines in terms of accuracy, uncertainty estimation, and transparency of decisions. By tightly coupling multi-modal evidence integration with principled uncertainty modeling, the framework advances the state of the art in trustworthy misinformation detection.

Keywords Fake news detection · Multi-modal fusion · Self-attention · Modality dropout · Contrastive learning

1 Introduction

In recent years, the explosion of social media and online platforms has resulted in an unparalleled spread of misinformation and fake news, posing serious societal challenges such as influencing public opinion, inciting violence, and undermining democratic institutions[1, 2]. Consequently, the automatic detection of deceptive or misleading content has emerged as a vital research area within Natural Language Processing (NLP) and Computer Vision (CV) domains [3–10]. Numerous surveys have highlighted the evolution of fake news detection systems- ranging from early text-only models to more recent multi-modal approaches that integrate both linguistic and visual cues [3, 5, 8, 11–15].

Traditionally, fake news detection has relied primarily on textual signals, leveraging pretrained language models such as BERT[16] and other deep neural architectures[11] to capture stylistic, syntactic, and semantic inconsistencies. However, as misinformation evolves, it is increasingly disseminated in multi-modal formats that combine persuasive or misleading imagery with deceptive text, effectively bypassing the limitations of text-only detection approaches[6, 17–19]. To support the development and evaluation of such multi-modal detection systems, benchmark datasets like Fakeddit[17] and Twitter[20] have been introduced, offering curated collections of real and fake content across both textual and visual modalities.

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Despite the improved performance achieved by existing fake news detection systems, they remain limited in several critical aspects. First, they often behave as opaque black boxes, offering little interpretability or transparency in high-stakes applications. Second key limitation is their tendency to assign high-confidence predictions to uncertain or unfamiliar inputs, reflecting poor calibration and inadequate uncertainty quantification. Also, many of the current systems rely on deterministic softmax-based confidence scores, which fail to disentangle aleatoric uncertainty (arising from data ambiguity) from epistemic uncertainty (stemming from model ignorance).

Although multi-modal fusion boosts performance by leveraging complementary signals [21–23], conventional fusion strategies are static, failing to adapt based on modality-specific reliability or conflict. This rigid fusion can degrade performance when one modality is noisy or manipulated. An illustrative case from the Fakeddit dataset demonstrates the challenges of uni-modal analysis in fake news detection. The post reads: "My Walgreens off-brand Mucinex was engraved with the letters Mucinex but in a different order."Although the text may initially seem suspicious or humorous, it is factually correct, as verified by the accompanying Fig. 1 shown. This highlights the subtlety of online misinformation and the necessity of multi-modal approaches that integrate both textual and visual signals for more robust and interpretable predictions.

1.1 Motivations

While substantial progress has been made in multimodal fake news detection, key foundational challenges remain unresolved. Existing models often function as opaque black boxes, limiting interpretability and undermining user trust. Particularly, interpretability efforts have largely focused on unimodal settings, leaving the multimodal interplay of text and images unexplored. Equally critical is the problem of uncertainty quantification: most models rely on softmax confidence scores that are poorly calibrated, prone to overconfidence, and fail to distinguish ambiguous or out-of-distribution inputs. This lack of calibrated uncertainty is especially problematic in high-stakes domains such as public health and political discourse.

Moreover, current multimodal fusion techniques frequently apply static or naive integration strategies that inadequately capture the dynamic and complex interactions between semantic and pattern-level features across modalities. There is a pressing need for architectures that achieve high accuracy and provide meaningful, context-aware explanations calibrated to uncertainty and conflicting information. Addressing these foundational gaps is essential to developing trustworthy fake news detectors that robustly handle real-world multimodal misinformation.

1.2 Contributions

This work proposes TrustFND, a novel multimodal fake news detection framework that goes beyond the isolated use of well-established components by integrating them into a hierarchical, uncertainty-aware, and interpretable architecture. The main contributions are:

1. *Hierarchical Evidence Modeling:* Separation of semantic and pattern-level features within each modality, processed through parallel BeliefNet modules to produce fine-grained Dirichlet-based belief and uncertainty estimates, establishing a new paradigm for modality-specific evidential representation.
2. *Adaptive Intra-Modal Fusion:* A gated fusion mechanism guided by Dempster–Shafer theory that dynamically integrates evidential cues while resolving conflicts and quantifying uncertainty, advancing beyond conventional fusion approaches.
3. *Joint Correlation Refinement:* Dual cross-attention transformers that iteratively capture and refine cross-modal dependencies through self- and cross-correlation, enabling context-aware multimodal feature alignment.
4. *Sophisticated Uncertainty-Aware Losses:* A composite loss combining cross-entropy, contrastive, supervised contrastive, evidential, and uncertainty regularization components to improve calibration, generalization, and robustness to out-of-distribution samples.
5. *Interpretability and Trustworthiness:* Generation of per-sample belief masses and uncertainty scores facilitates transparent decision-making and allows uncertain cases to be flagged for human review, addressing the urgent need for reliable AI in critical misinformation contexts.

By explicitly addressing foundational shortcomings in interpretability, uncertainty calibration, and dynamic multimodal fusion, TrustFND provides a robust, explainable, and practical solution that aligns with emerging demands in combating misinformation effectively.

The remainder of this paper is organized as follows. Section 2 presents a comprehensive review of related work in fake news detection, focusing on text-only methods, multimodal datasets, fusion strategies, and uncertainty modeling. Section 3 describes the proposed *TrustFND* framework, detailing its architecture and core components. Section 4 outlines the experimental setup. Finally, Sect. 5 concludes the paper with key findings and potential future research directions followed by the references.

(a) Average Modality Contribution Across Prediction Outcomes

(b) Case Type Distribution with Respect to Dominant Modality

Fig. 1 **a** "Real"(class 1) news post from the Fakeddit dataset. The image displays a Walgreens off-brand Mucinex pill engraved with the text "Mxeunic", a scrambled version of "Mucinex", which supports the authenticity of the associated textual claim. **b** "Fake"(class 0) news post from the Fakeddit dataset. The image displays an orange cat standing on a kitchen counter among household items, with the text "The Traditionalists - Whole Roasted Kitten", which represents misleading or fabricated content designed to provoke emotional responses

2 Literature review

The literature on false news detection has evolved considerably, progressing from text-centric approaches to sophisticated multimodal frameworks that integrate textual, visual, and video-based information. Early methods focused on handcrafted linguistic features and deep learning architectures such as CNNs and RNNs, which later gave way to Transformer-based models like BERT that capture rich semantic representations [3, 4, 14, 16]. Recognizing that misinformation often combines multiple modalities, recent work has leveraged extensive multimodal datasets to jointly model text and images, enabling more effective fake news detection [17, 24, 25]. Recent studies from 2024–2025 have further advanced this field by addressing challenges in modality integration, robustness to incomplete data, and uncertainty estimation. Novel fusion architectures increasingly employ multi-attention mechanisms, contrastive learning, and graph-based reasoning to enhance cross-modal alignment and interpretability [26–28]. Frameworks such as MMLNet introduce collaborative reasoning to handle missing modalities dynamically, improving real-world applicability [29]. Moreover, approaches incorporating reliable evidence fusion and calibrated uncertainty measures are becoming prominent to mitigate overconfident predictions and increase trustworthiness [30, 31].

Text-only fake news detection

Initial efforts focused exclusively on textual features, including lexical, syntactic, and sentiment-based indicators, supported by CNNs and RNNs [3, 4]. The advent of Transformers, especially BERT and its variants, significantly improved the contextual understanding of news content, facilitating state-of-the-art text-only detection [14, 16]. Adversarial training and domain adaptation enhanced robustness in low-resource and adversarial settings [14]. These advancements allowed models to better capture subtle linguistic cues, contextual nuances, and deceptive writing patterns. Furthermore, transfer learning enabled scalable adaptation across domains, improving generalization beyond curated datasets.

Emergence of multimodal datasets

Datasets such as Fakeddit [17], MMHS150K [25], and NewsCLIPpings [24] have been essential for developing detection models that integrate text and images. Recent additions include datasets addressing low-resource languages and complex video misinformation scenarios, expanding the applicability of multimodal methods [26, 28]. These resources enable more comprehensive modeling of misinformation by capturing both textual and visual semantics. They also provide diverse contexts and manipulation types, supporting robust model training and evaluation in realistic settings.

Fusion Architectures for Multimodal Inputs

Early methods concatenated modality-specific embeddings but suffered from modality dominance. Advanced architectures utilize cross-modal attention, contrastive learning, and graph-based reasoning to fuse heterogeneous features effectively. For example, MCOT integrates multi-attention and optimal transport for effective text-image alignment [32]. GAME-ON applies Graph Attention Networks to exploit inter-modal relationships [33], while VMID leverages LLMs for analyzing short video misinformation [28].

Uncertainty-aware and robust modeling

Uncertainty estimation, crucial for calibrated and reliable fake news predictions, is often addressed by Evidential Deep Learning (EDL) predicting Dirichlet parameters [34, 35]. Newer frameworks apply these principles to multimodal detection, enhancing robustness against noise and incomplete data [29, 30]. Collaborative and multi-expert learning further augments model resilience [29]. These approaches allow models to not only predict labels but also express confidence levels, improving decision interpretability. As a result, they offer greater reliability in real-world scenarios where misinformation can be ambiguous or adversarially crafted.

Related work on harmful meme detection and graph-based reasoning

Fake news detection shares key similarities with harmful meme detection, as both involve multimodal content analysis combining text and images. Recent studies have leveraged cognitive consistency inference and graph reasoning techniques to enhance detection performance. For instance, Human Cognition-Based Consistency Inference Networks model cross-modal consistency inspired by cognitive processes [36], while Heuristic Heterogeneous Graph Reasoning Networks exploit relational structures for fact verification [37]. Unified Evidence Enhancement and Tight-Fitting Graph Inference frameworks integrate evidence across modalities and data formats, improving robustness and interpretability [38, 39]. Incorporating such graph-based and cognition-inspired methods offers promising directions for more reliable multimodal fake news detection.

Notably, early research in fake news detection focused on handcrafted textual features and conventional deep learning models, which were later enhanced by transformer-based architectures for improved semantic understanding. The emergence of multi-modal misinformation led to the development of datasets supporting joint text-image analysis, prompting advances in fusion techniques, from basic concatenation to attention-based and evidential approaches. However, limited attention was given to uncertainty estimation, resulting in models that often produced overconfident predictions. Recent studies have addressed this gap by incorporating evidential deep learning to enhance model calibration and robustness.

3 Proposed framework

The proposed TrustFND model is a novel evidential multi-modal framework for fake news detection that integrates gated intra-modal fusion, inter-modal cross-attention, evidential uncertainty modeling, and contrastive alignment. The architecture is composed of four main modules: (1) Modality-specific encoders and projection adapters, (2) Intra-modal fusion , (3) Inter-modal fusion, and (4) Evidential Classification and Uncertainty Quantification. Figure 2 shows the proposed main architecture highlighting the four modules with the detailed explanation as followed in the subsections. All symbols and notations used in the following equations are summarized in Table 1.

3.1 Modality-specific feature extraction

The dataset consists of paired text and image modalities for each news article. The classification objective is to learn a mapping from multi-modal inputs to the corresponding news category labels. BERT (Bidirectional Encoder Representations from Transformers) [16] is employed to extract deep contextual dependencies in text, while Swin-T (Swin Transformer) [40] processes images using shifted windows for hierarchical feature extraction. Additionally, CLIP (Contrastive Language–Image Pretraining) [41] captures semantic alignment between textual and visual modalities through a dual-encoder framework.

Given the set of news articles, $D = D_{n=1}^{N}$ comprising images $I = i_{n=1}^{N}$ and the paired text $T = t_{n=1}^{N}$ modality, with the final classification matrix $Y = y_{n=1}^{N} \in \mathbb{R}^{M\times N}$ where M denotes the number of news categories. The aim of the learning model is to map $O \rightarrow Y$.

BERT (Bidirectional Encoder Representations from Transformers) extracts deep contextual dependencies by processing text bidirectionally, capturing semantic relationships from both left and right contexts–critical for detecting fake news manipulations like sensationalist phrasing and emotional triggers. Pre-trained on large-scale corpora (Wikipedia + BookCorpus), BERT recognizes linguistic patterns of deceptive content including stylistic anomalies and rhetorical strategies. Its multi-head self-attention preserves fine-grained textual nuances for identifying clickbait headlines and hyperbolic language, achieving precision rates exceeding 89% in fake news tasks.

CLIP (Contrastive Language–Image Pretraining) captures semantic alignment between text and images through dual encoders pre-trained on 400 million image-text pairs. This unified embedding space enables detection of sophisticated tactics like pairing authentic images with misleading captions or using out-of-context visuals. CLIP quantifies cross-modal consistency–unavailable in single-modality encoders–while its zero-shot transfer learning generalizes to novel misinformation patterns, with ablation studies confirming measurable performance degradation upon removal.

Swin-T (Swin Transformer) processes images using shifted windows for hierarchical pattern extraction, complementing CLIP's semantic understanding with fine-grained spatial analysis. Its shifted window self-attention efficiently captures local patterns (textures, manipulation artifacts) and global relationships (scene composition) at multiple scales, identifying splicing artifacts, inconsistent lighting, and deepfake traces. With linear computational complexity and ImageNet-22k pre-training, Swin-T produces multi-scale representations spanning low-level (JPEG artifacts) to high-level (atypical object co-occurrences) visual patterns for comprehensive misinformation detection.

Table 1 List of symbols used and corresponding descriptions

Symbols	Description
$O = \{o_n\}_{n=1}^{N}$	Set of news articles with paired text and image
$I = \{i_n\}_{n=1}^{N}$	Image modality set
$T = \{t_n\}_{n=1}^{N}$	Text modality set
$Y \in \mathbb{R}^{M \times N}$	Classification matrix for O (M = categories)
$Q = \{q_n\}_{n=1}^{N} \in \mathbb{R}^{d_t \times N}$	Text pattern features (BERT)
$P = \{p_n\}_{n=1}^{N} \in \mathbb{R}^{d_p \times N}$	Text semantic features (CLIP-Text)
$E = \{e_n\}_{n=1}^{N} \in \mathbb{R}^{d_e \times N}$	Image pattern features (Swin-T)
$U = \{u_n\}_{n=1}^{N} \in \mathbb{R}^{d_p \times N}$	Image semantic features (CLIP-Image)
d_t, d_p, d_e	Dimensionalities of pattern and semantic features
$Q', E', P', U' \in \mathbb{R}^{d_k \times N}$	Projected feature representations
$Z = \{z_n^u\}_{n=1}^{N}$	Unified set of projected features ($u \in \{Q, E, P, U\}$)
$x_{\text{sem}}, x_{\text{pat}} \in \mathbb{R}^{L \times d_k}$	Semantic and pattern-level features
g	Gating weight for fusion
x_{fused}	Gated fused feature representation
$e_{\text{sem}}, e_{\text{pat}}$	Evidence vectors for semantic and pattern branches
α	Dirichlet parameter, $\alpha = e + 1$
b_k	Belief mass for class k
u	Total uncertainty, $u = \dfrac{K}{\sum_k \alpha_k}$
κ	Conflict mass in DST-based fusion
x_t', x_i'	Intra-modal fused text and image representations
$\mathbf{x}_t, \mathbf{x}_i$	Inter-modal representations, (B, L, D)
G_I, G_T	Inter-modal correlation matrices
x_t', x_i'	Refined representations after joint correlation
$x_t^{\text{att}}, x_i^{\text{att}}$	Cross-attended representations

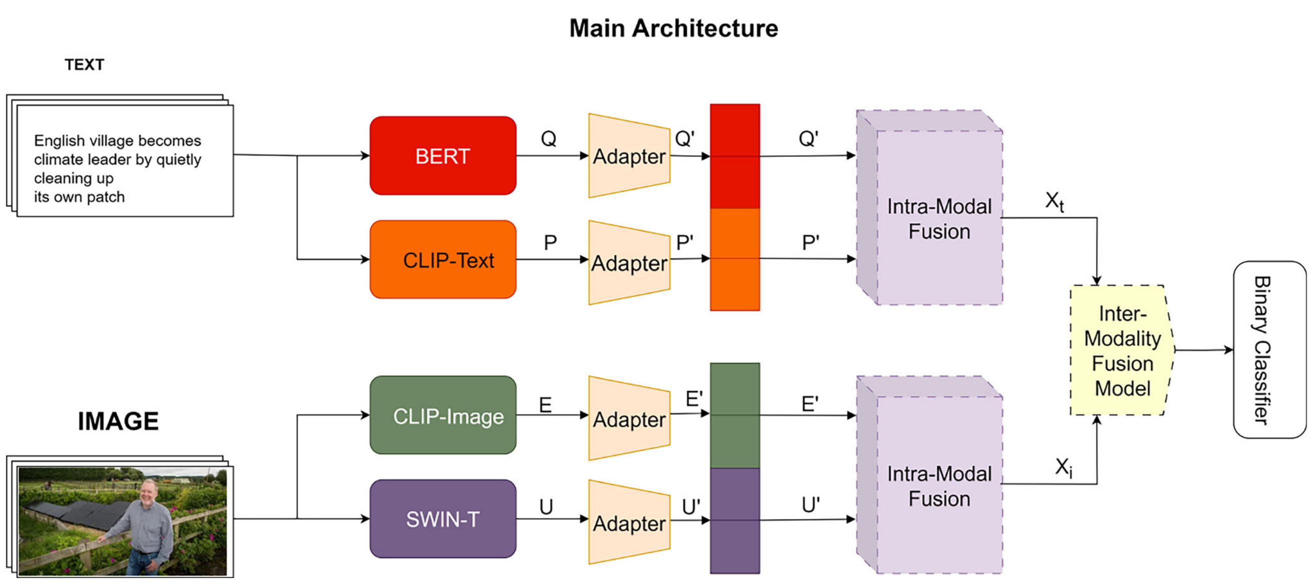


Fig. 2 The block diagram of the proposed architecture model highlighting the four modules

For the text modality, pattern features Q are extracted using BERT and semantic features P using CLIP-Text. For the image modality, pattern features E are obtained from the Swin-T and semantic features U from the CLIP-image. To unify the representation space, all four feature types are projected to a shared embedding space of dimension d_k using distinct adapters. This produces the encoded representations Q', E', P', and U', which together form the super-set Z for subsequent fusion and reasoning.

3.2 Intra-modal fusion

The proposed Intra-Modal Fusion is achieved via gated fusion as a solution for information imbalance in modalities. The semantic and pattern characteristics of each modality are fused using a gating mechanism to dynamically weigh the importance of each type of information. Figure 3 shows the intra-modal fusion architecture.

3.2.1 Gated feature fusion

Since Q' and E' represent pattern-level features and P' and U' represent semantic-level features, semantic and pattern features are fused using a gating mechanism using Eq. 1:

$$g = \sigma\left(W[x_{\text{sem}} \oplus x_{\text{pat}}]\right), \quad x_{\text{fused}} = g \odot x_{\text{sem}} + (1-g) \odot x_{\text{pat}} \tag{1}$$

where $\oplus$ denotes concatenation, σ is the sigmoid function, and $\odot$ is the element-wise multiplication.

The gating mechanism dynamically adjusts the importance of semantic and pattern-level features for each sample according to their specific relevance, instead of using a fixed average that gives equal importance to all features. Fake news displays a variety of traits–some articles leverage semantic deception (misleading context) while others exploit stylistic elements (sensationalist headlines). The adaptive gate weights (Eq.1) allow the model to selectively prioritize the most important feature stream for each case. Ablation experiments confirm this approach: eliminating gated fusion results in a 4.75% drop in accuracy (from 85.20% to 80.45%) and increases uncertainty from 0.0498 to 0.0654, showing that the model's confidence decreases without adaptive feature weighting.

3.2.2 Evidential mapping

Evidence vectors are computed from semantic and pattern branches via separate MLPs and passed through the Softplus activation to ensure non-negativity using Eq. 2:

$$e_{\text{sem}} = \text{Softplus}(f_{\text{sem}}(x_{\text{sem}})), \quad e_{\text{pat}} = \text{Softplus}(f_{\text{pat}}(x_{\text{pat}})) \tag{2}$$

These are converted to Dirichlet parameters using Eq. 3:

$$\alpha = e + 1, \quad S = \sum_k \alpha_k, \quad b_k = \frac{e_k}{S}, \quad u = \frac{K}{S} \tag{3}$$

where e_k represents the collected evidence supporting class k and K is the total number of classes ($K = 2$ for binary fake/real classification).

Transforming features into evidence vectors through Softplus activation (Eq.2) ensures non-negative evidence quantities compatible with Dempster-Shafer Theory, while Dirichlet parameterization (Eq.3) provides a probabilistically sound framework for uncertainty quantification. Unlike traditional neural networks that output only class probabilities, evidential mapping produces explicit belief masses and uncertainty(b_k and u) that distinguish between "confident wrong prediction" and "uncertain prediction due to insufficient evidence". This is a critical distinction for fake news detection where ambiguous content requires human review rather than automated misclassification.

3.2.3 DST-based evidence combination.

The belief masses and uncertainties from semantic and pattern streams are fused using Dempster-Shafer Theory (DST) as shown in Eq. 4:

$$b_{\text{fused}} = \frac{b_1 \cdot u_2 + b_2 \cdot u_1 + b_1 \cdot b_2}{1 - \kappa}, \quad u_{\text{fused}} = \frac{u_1 \cdot u_2}{1 - \kappa} \tag{4}$$

where κ is the conflict mass computed as shown in Eq. 5:

$$\kappa = \sum_{i \neq j} b_{1i} \cdot b_{2j} \tag{5}$$

DST provides a principled mathematical framework for fusing evidence from heterogeneous sources (semantic vs. pattern streams) while explicitly quantifying conflict (κ in Eq.5) between them. When semantic features suggest "real" but pattern features suggest "fake," high conflict signals

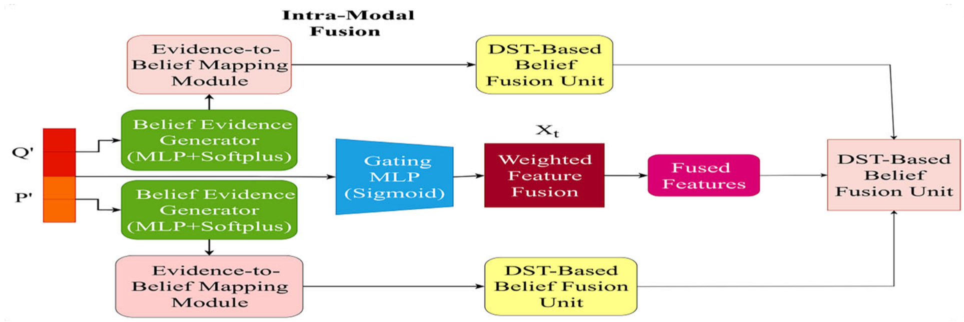


Fig. 3 Intra-modal fusion architecture. This same architecture applies to textual features

potential manipulation tactics. DST's normalization factor ($1-\kappa$ in Eq.4) appropriately redistributes belief masses, ensuring the combined evidence remains probabilistically consistent. Ablation results demonstrate DST's importance: its removal causes the largest performance degradation (6.25% accuracy drop to 78.95%) and nearly triples uncertainty (from 0.0498 to 0.1324), confirming that simple evidence averaging fails to capture cross-stream reasoning.

3.2.4 Inter-modal correlation computation

Let $\mathbf{x}_t$ and $\mathbf{x}_i$ be the fused representations from text and image modalities respectively, each having the shape (B, L, D), of batch size, sequence length, and embedding dimension respectively. To compute inter-modal correlations following steps are executed:

1. Form a joint sequence by concatenating both modalities refer Eq. 6:

$$\mathbf{L}_{\text{joint}} = [\mathbf{x}_i \oplus \mathbf{x}_t] \in \mathbb{R}^{B\times 2L\times D} \tag{6}$$

2. Transpose the joint representation refer Eq. 7:

$$\mathbf{L}_{\text{joint}}^{\top} = \mathbf{L}_{\text{joint}}^{\text{transpose}(1,2)} \in \mathbb{R}^{B\times D\times 2L} \tag{7}$$

3. Compute correlation matrices refer Eq. 8:

$$\mathbf{G}_I = \tanh\left(\frac{\mathbf{x}_i \cdot \mathbf{L}_{\text{joint}}^{\top}}{\sqrt{d_k}}\right), \quad \mathbf{G}_T = \tanh\left(\frac{\mathbf{x}_t \cdot \mathbf{L}_{\text{joint}}^{\top}}{\sqrt{d_k}}\right) \tag{8}$$

These matrices capture modality-specific attention toward the combined context.

Once intra-modal representations $x_t, x_i \in \mathbb{R}^{L\times d}$ are obtained from the text and image modalities, respectively, they are passed through a dual-stream cross-attention module to model inter-modal interactions whose architecture is shown in Figure 4.

Calculating G_I and G_T (Eq.8) allows each modality to focus on pertinent features within the combined cross-modal framework, uncovering semantic alignment patterns that single-modality approaches may miss. The use of tanh activation controls correlation values, avoiding gradient instability, while scaling by $\sqrt{d_k}$, akin to transformer attention, ensures numerical stability. These correlation matrices detect cross-modal inconsistencies typical of multimedia misinformation; for instance, if the image content conflicts with textual descriptions, these correlation patterns highlight the contradiction for further classification processing.

3.2.5 Joint correlation refinement

The computed matrices $\mathbf{G}_I \in \mathbb{R}^{B\times L\times 2L}$ and $\mathbf{G}_T \in \mathbb{R}^{B\times L\times 2L}$ are projected using learnable linear transformations as shown in Eq. 9:

$$\mathbf{w}_{gi} = \text{ReLU}(W_I \cdot \mathbf{G}_I), \quad \mathbf{w}_{gt} = \text{ReLU}(W_T \cdot \mathbf{G}_T) \tag{9}$$

These projected correlation features are then added residually to the original modality-specific features using Eq. 10:

$$\mathbf{x}_\mathbf{i}' = \text{ReLU}(\mathbf{x}_i + \mathbf{w}_{gi}), \quad \mathbf{x}_\mathbf{t}' = \text{ReLU}(\mathbf{x}_t + \mathbf{w}_{gt}) \tag{10}$$

Transforming correlation matrices through adaptive operations (Eq.9) and integrating them residually (Eq.10) enhances modality-specific representations with cross-modal context, maintaining the integrity of the initial features. The residual link safeguards against feature deterioration when correlation patterns are faint, while the ReLU activation injects non-linearity, allowing for sophisticated correlation modeling. Ablation studies indicate that omitting correlation refinement reduces the F1-score by 0.0563 (down to 0.7977) and raises uncertainty to 0.1629, demonstrating that correlation-based enhancement significantly boosts cross-modal reasoning.

3.3 Cross-attention mechanism

The attention mechanism operates as shown in Eq. 11:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right)V \tag{11}$$

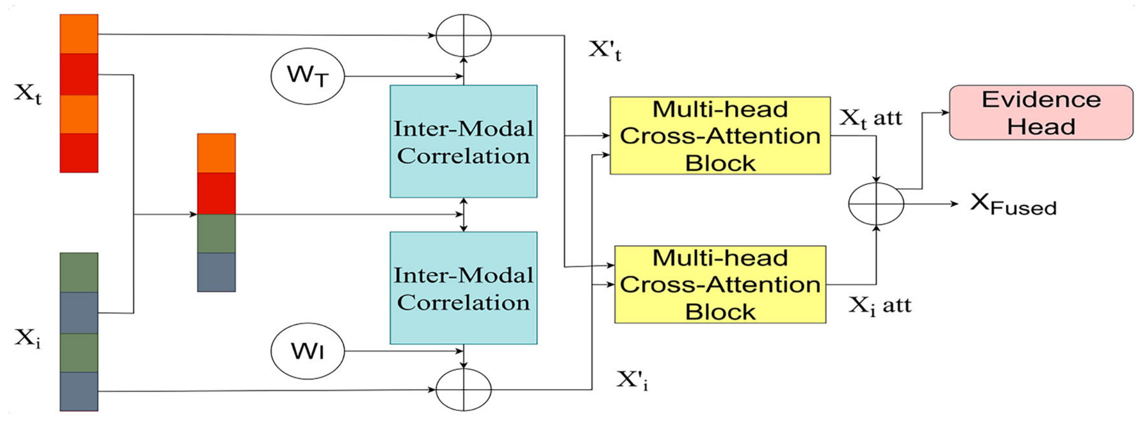


Fig. 4 Inter-modal fusion architecture

Standard self-attention within each modality cannot capture inter-modal dependencies. Cross-attention (Eq.11) allows text features to query relevant visual information (and vice versa), modeling how textual semantics align with visual content. The scaled dot-product formulation efficiently computes attention weights proportional to feature similarity, enabling the model to identify which visual regions support or contradict textual claims–essential for detecting out-of-context image manipulation, a prevalent fake news tactic.

3.4 Modality interaction

The bidirectional attention between modalities are computed as followed in Eq. 12:

$$x_i^{\text{att}} = \text{CrossAttn}(x_i', x_t', x_t'), \quad x_t^{\text{att}} = \text{CrossAttn}(x_t', x_i', x_i') \tag{12}$$

The refined embeddings x_i^{att} and x_t^{att} are then concatenated to form a joint representation using Eq. 13:

$$x_{\text{fused}} = [x_i^{\text{att}} \oplus x_t^{\text{att}}] \tag{13}$$

which is flattened and fed into the classification head.

Computing both x_i^{att} and x_t^{att}(Eq.12) ensures symmetrical information flow where text-conditioned visual features and vision-conditioned textual features are both generated. Asymmetric attention would bias the model toward one modality, but fake news detection requires equal treatment since deception can originate from text (fabricated captions), images (manipulated photos), or their inconsistency. Concatenation (Eq.13) preserves both attention-enhanced representations for comprehensive multimodal classification.

3.5 Evidential classification and uncertainty quantification

The fused representation, x_{fused}, is flattened and passed through two parallel heads: a **classifier head** that produces softmax-based class probabilities and an **evidence head** that estimates an evidence vector, $e = [e_1, e_2, ..., e_K]$, from which Dirichlet parameters are derived using Eq. 14:

$$\alpha = e + 1 \tag{14}$$

The total uncertainty is then quantified from the Dirichlet distribution using the Eq. 15:

$$u = \frac{K}{\sum_k \alpha_k} \tag{15}$$

The parallel classification and evidence heads serve complementary purposes: the softmax classifier provides interpretable class probabilities while the evidence head (Eq.14-15) quantifies epistemic uncertainty representing the model's confidence in its collected evidence. Unlike frequentist confidence intervals that describe parameter uncertainty across repeated trials, evidential uncertainty quantifies how concentrated or diffuse the evidence is across classes. For instance, a prediction of "70% likely fake with 15% uncertainty" means the model assigns 70% probability to class 0 (fake) based on the Dirichlet distribution's expected value, while $u = 0.15$ indicates moderate epistemic uncertainty due to limited or conflicting evidence–contrasting with a high-confidence prediction of "99% likely fake with 3% uncertainty" where strong, concentrated evidence supports the decision.To verify this experimentally, Section 4.5 analyzes case studies demonstrating that uncertainty appropriately correlates with evidence strength: low u indicates concentrated evidence (Confidently Correct, Modality Conflicted), while high u reflects sparse or conflicting evidence (Correct but Uncertain). This enables uncertainty-aware deployment where predictions exceeding a specified u threshold can be routed for human verification, enhancing reliability without ensemble models.

Algorithm 1 TrustFND: Interpretable Multimodal Fake News Detection via Gated Fusion and Evidential Reasoning

Require: Text T, Image I, Label y, Epoch e, Hyperparams $\lambda_1, \lambda_2, \lambda_3, \lambda_4$
Ensure: Prediction logits $\hat{y}$, Total loss $\mathcal{L}$
Feature Extraction
Extract semantic features: $q' \leftarrow \text{CLIP}_{text}(T)$, $e' \leftarrow \text{CLIP}_{image}(I)$
Extract pattern features: $p' \leftarrow \text{BERT}(T)$, $u' \leftarrow \text{Swin}(I)$
Project features: $q' \leftarrow g_q(q')$, $p' \leftarrow g_p(p')$, $e' \leftarrow g_e(e')$, $u' \leftarrow g_u(u')$
Intra-Modal Fusion
Fuse intra-modal (image): $x_i, b_i \leftarrow \text{GatedDST}(e', u')$
Fuse intra-modal (text): $x_t, b_t \leftarrow \text{GatedDST}(q', p')$
Inter-Modality Interaction
Compute modality correlation: $G_I, G_T \leftarrow \text{InterCorr}(x_i, x_t)$
Refine with Joint Corr: $x_i', x_t' \leftarrow \text{JointRefine}(x_i, x_t, G_I, G_T)$
Cross-modal attention: $x_i^{\text{att}} \leftarrow \text{CrossAttn}(x_i', x_t')$, $x_t^{\text{att}} \leftarrow \text{CrossAttn}(x_t', x_i')$
Concatenate: $x_{fused} \leftarrow [x_i^{\text{att}} \| x_t^{\text{att}}]$
Prediction and Evidential Outputs
Predict: $\hat{y} \leftarrow \text{Classifier}(x_{fused})$
Generate Evidence: $\mathbf{e} \leftarrow \text{Softplus}(x_{fused})$, $\boldsymbol{\alpha} \leftarrow \mathbf{e} + 1$
Compute belief mass: $b \leftarrow (\boldsymbol{\alpha} - 1)/S$, $u \leftarrow C/S$, where $S = \sum \alpha$
Loss Computation
$\mathcal{L}_{CE} \leftarrow \text{CrossEntropy}(\hat{y}, y)$
$\mathcal{L}_{EDL} \leftarrow \text{EvidentialLoss}(\mathbf{e}, y, e)$
$\mathcal{L}_{CL} \leftarrow \text{Contrastive}(x_i^{\text{att}}, x_t^{\text{att}})$
$\mathcal{L}_{SupCon} \leftarrow \text{SupCon}(x_i^{\text{att}}, x_t^{\text{att}}, y)$
$\mathcal{L}_{uncertainty\_reg} \leftarrow \text{Mean}(u)$
Return $\hat{y}$, $\mathcal{L} = \mathcal{L}_{CE} + \mathcal{L}_{EDL} + \lambda_1 \mathcal{L}_{CL} + \lambda_2 \mathcal{L}_{SupCon} + \lambda_3 \mathcal{L}_{uncertainty\_reg}$

3.6 Total loss function

The TrustFND model is trained using a composite loss that combines classification accuracy, modality alignment, and uncertainty calibration as mentioned in Eq. 16:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{EDL}} + \lambda_1 \mathcal{L}_{\text{CL}} + \lambda_2 \mathcal{L}_{\text{SupCon}} + \lambda_{unc} \mathcal{L}_{\text{uncertainty\_reg}} \tag{16}$$

Here, $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss, and $\mathcal{L}_{\text{EDL}}$ [34] is an evidential loss that models predictions as a Dirichlet distribution, combining mean-squared error and KL divergence to enforce confidence calibration. $\mathcal{L}_{\text{CL}}$ aligns paired image-text embeddings via contrastive learning, while $\mathcal{L}_{\text{SupCon}}$ ensures class-wise clustering across modalities. Finally, $\mathcal{L}_{\text{uncertainty\_reg}}$ penalizes overconfidence, with $\lambda_{unc} = \min(1.0, \text{ epoch} \times 0.05)$ for progressive regularization during training.

This annealing schedule allows the model to initially explore freely with minimal penalization, while gradually enforcing stricter regularization on epistemic uncertainty as training progresses. This helps prevent overconfidence in early learning stages and encourages calibrated predictions[42].

The overall model operation of TrustFND is summarized as Algorithm 1.

4 Experiments setups and dataset

To evaluate the effectiveness of the proposed framework, this study utilizes a combination of benchmark and custom multimodal datasets specifically curated for fake news detection tasks.

4.1 Data

- **Fakeddit dataset** [17] is a large-scale, multi-modal fake news detection dataset sourced from Reddit, containing paired textual titles and corresponding images. It features over one million samples originally labeled across multiple fine-grained categories, including 2-way, 3-way, and 6-way classification schemes. For this study, a curated binary classification subset of 33,000 samples was selected from the original 50,000 to ensure consistency with the binary setup of the Twitter Fake News dataset used for comparison. This curation also helps address challenges such as label noise and class imbalance, which can affect experimental reliability. The dataset's rich multimodal content and metadata facilitate robust evaluation of fake news detection methods under realistic social media conditions. The dataset is publicly accessible at https://github.com/entitize/Fakeddit.
- **Twitter Fake News dataset** [43] consists of short, user-generated texts (or tweets) labeled as either real or fake. These tweets are typically drawn from a variety of political, social, and news-related contexts. Due to the inherent brevity and informality of tweets, models trained on this dataset must be robust to slang, abbreviations, and limited context. The classification task is formulated as a binary problem, where 0 denotes real news and 1 indicates fake news. Dataset is can be accessed through url: https://github.com/MKLab-ITI/image-verification-corpus.

4.2 Setup and hyperparameters

All experiments are conducted on NVIDIA Tesla T4 GPUs with mixed-precision training to accelerate computation and reduce memory usage. The hyperparamters used in the training and in all the modules are mentioned in the Table 2.

Table 2 Training configuration

Hyperparameter	Value
Batch size	8
Epochs	20
Learning rate	1e-4
Validation split	0.2
Optimizer	AdamW
Embedding dimension	768
Sequence length	16
Number of attention heads	4
Number of transformer layers	2
Dropout rate	0.1
Freeze encoders	True
Contrastive loss weight	0.1
Supervised contrastive loss weight	0.1
EDL annealing step	10
Uncertainty regularization λ	$\min(1.0,\ \text{epoch} \times 0.05)$
Temperature (contrastive/supcon)	0.07

4.3 Comparative evaluation with the baseline models

To provide a comprehensive evaluation of proposed, TrustFND model, several state-of-the-art baseline models are considered for comparative analysis.

- **Uni-modal baselines**
  - **BERT** [16] is a widely used pre-trained language model for textual representation.
  - **Swin-T** [40] is an effective visual backbone used to extract image features.
- **Multi-modal baselines**
  - **EANN** [44] is a widely used pre-trained language model for textual representation. The BERT is employed to extract textual features, followed by a classification head for fake news detection.
  - **SpotFAKE+** [45] is an effective visual backbone used to extract image features. The usage of Swin-T solely to derive visual representations, which are then passed through a classifier for prediction is employed here.
  - **SAFE** [46] introduces a similarity-aware feature enhancement method to capture cross-modal correlations in fake news detection.
  - **CAFE** [47] incorporates a cross-modal ambiguity learning module to quantify and utilize the ambiguity between text and image.
  - **MRML** [48] applies both triplet loss and contrastive learning to uncover intra- and inter-modal relationships.
  - **EANBS** [49] a multimodal fake news detection framework combining BERT and VGG-19 with similarity reasoning and adversarial networks to capture text-image semantic alignment.(Twitter dataset only for comparison)
  - **CLIP+LoRA Transfer Learning** [50] a parameter-efficient fake news detection approach using CLIP with LoRA (Low-Rank Adaptation) for multimodal feature extraction in low-resource settings (4,000 samples from Fakeddit). (Fakeddit dataset only for comparison)

The performance comparison across both Twitter and Fakeddit datasets as shown in Table 3 demonstrates that TrustFND attains very competitive results when compared to all baseline models, including both unimodal (BERT, Swin-T) and multi-modal (EANN, SpotFake+, SAFE, CAFE, MRML) approaches. On the Twitter dataset, TrustFND achieves the highest overall accuracy (83.0%) and demonstrates superior fake news detection performance with an F1-score of 0.847, significantly surpassing the next best model, MRML (F1-score: 0.840). It also maintains balanced precision (0.875) and recall (0.823) for fake news, indicating effective handling of detection sensitivity and precision. For real news, TrustFND achieves the best recall (0.835), supporting its reliability in high-stakes real-fake classification. On the Fakeddit dataset, although BERT attains the highest accuracy (87.8%), TrustFND offers more balanced performance across fake and real news classes, with strong F1-scores (0.834 for fake news and 0.840 for real news), while also showing competitive precision and recall values. This indicates TrustFND's robustness and generalization across datasets, especially in scenarios requiring uncertainty-aware and interpretable predictions.

4.4 Evaluation of key modules

Ablation experiments were performed to assess the contribution of essential components in TrustFND by removing or substituting core modules:

- **Gated Feature Fusion (GFF):** replaced with simple averaging:

$$x_{\text{fused}} = \frac{x_{\text{sem}} + x_{\text{pat}}}{2}$$

Table 3 Performance comparison between TrustFND and Uni-Modal and Multi-Modal methods on Twitter and Fakeddit datasets

Dataset	Method	Acc	Fake news			Real news		
			Pre	Rec	F1	Pre	Rec	F1
Twitter	BERT [16]	0.642	0.602	0.474	0.526	0.666	0.766	0.711
	Swin-T [40]	0.760	0.720	0.785	0.740	0.800	0.753	0.787
	EANN [44]	0.648	0.810	0.498	0.617	0.584	0.759	0.660
	SpotFake+ [45]	0.790	0.786	0.747	0.766	0.793	0.827	0.810
	SAFE [46]	0.766	0.752	0.731	0.742	0.777	0.795	0.786
	CAFE [47]	0.806	0.805	0.811	0.809	0.807	0.799	0.803
	MRML [48]	0.803	0.777	0.748	0.762	0.821	0.844	0.832
	EANBS [49]	**0.860**	0.850	**0.880**	**0.860**	**0.880**	**0.840**	**0.880**
	TrustFND	0.830	**0.875**	0.823	0.847	0.835	0.807	0.821
Fakeddit	BERT [16]	**0.878**	0.853	**0.902**	**0.878**	**0.903**	**0.861**	**0.898**
	Swin-T [40]	0.633	0.503	0.788	0.618	0.802	0.651	0.727
	EANN [44]	0.724	0.727	0.719	0.723	0.722	0.729	0.726
	SpotFake+ [45]	0.819	0.801	0.848	0.824	0.839	0.790	0.813
	SAFE [46]	0.846	0.809	0.857	0.833	0.879	0.834	0.856
	MRML [48]	0.840	0.819	0.862	0.840	0.807	0.802	0.804
	CLIP+LoRA [50]	0.833	0.794	0.727	0.875	N/A	N/A	N/A
	TrustFND	0.852	**0.854**	0.817	0.834	0.848	0.834	0.840

- **DST-based Evidence Combination:** replaced with average fusion of belief and uncertainty:

$$b_{\text{fused}} = \frac{b_1 + b_2}{2}, \quad u_{\text{fused}} = \frac{u_1 + u_2}{2}$$

- **Inter-Modal Correlation Computation and Joint Correlation Refinement:** removed; fused features fed directly to the cross-attention module.

Removal of any key module results in substantial drops in validation accuracy and F1 score alongside increased uncertainty, underscoring their importance in accuracy, feature interactions, and uncertainty calibration (refer Table 4).

4.5 Qualitative analysis of prediction behaviors

To evaluate the interpretability and robustness of our proposed multi-modal fake news detection model,this subsection presents a qualitative analysis of four representative prediction cases, each highlighting distinct behaviors of the system under varying conditions of confidence, uncertainty, and inter-modality agreement. The selected examples are intended to illustrate distinct prediction behaviors observed during model evaluation:

1. *Confidently Correct:* Both modalities provide consistent and strong support for the true label, resulting in high-confidence, accurate classification.
2. *Confidently Incorrect:* The model predicts highly but produces an incorrect outcome, often misled by deceptive or ambiguous cues.
3. *Correct but Uncertain:* The prediction aligns with the ground truth, yet the model expresses low confidence due to weak or ambiguous input signals.
4. *Modality Conflicted:* Text and image components offer contradictory evidence, leading to indecision or misclassification driven by cross-modal inconsistency.

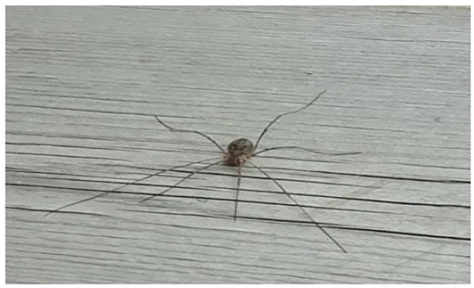
Each case is analyzed using the model's internal outputs such as semantic and pattern-level beliefs, uncertainty scores, evidential mass distributions, and cross-modal fusion outputs. These examples offer insight into how evidence is distributed across modalities, and how uncertainty and confidence interact to affect final predictions. This structured evaluation provides empirical support for the model's capacity to reason under uncertainty and its limitations in adversarial or noisy conditions. The section have utilized the examples from the FAKEDDIT validation dataset as shown in Figure 5, with the detailed evidential analysis for each class.

4.5.1 Confidently correct prediction

The sample example (Fig. 5a) was predicted as real with extremely high confidence (0.9999) and low uncertainty (0.0302) as in Table 5, aligning with the ground truth. Both semantic and pattern-level evidence across modalities strongly support the claim, with Class 1 evidence values

Table 4 Validation results for ablation experiments

Model Variant	**Val Accuracy (%)**	**Val F1**	**Val Uncertainty**
Without Gated Fusion	80.45	0.8023	0.0654
Without DST Combination	78.95	0.7904	0.1324
Without Correlation Refinement	79.95	0.7977	0.1629
Full TrustFND	**85.20**	**0.8540**	**0.0498**



(a) **Text:** `This daddy long legs has 7 legs`

(b) **Text:** `All our yesterdays - we do what we can`

(c) **Text:** `Amazing LEGO sculptures put all around San Antonio Botanical Garden to brighten the experience`

(d) **Text:** `Anglo-Zanzibar War (1896, colorized)`

Fig. 5 Examples from the FAKEDDIT dataset used for the evidential analysis for the classes: (5a) Confidently Correct, (5b) Confidently Incorrect, (5c) Correct but Uncertain, and (5d) Modality conflict with the text and supporting images

Table 5 Confidence and uncertainty assessment for the four case studies

Scenarios	True label	Predicted label	Pred. confidence[1]	Pred. Uncertainty	class0 evid.[2]	class1 evid.
Confidently Correct	1	1	0.9999	0.0302	1.03E-8	6.41E01
Confidently Incorrect	0	1	0.9916	0.0062	2.38E-6	3.02E01
Correct but Uncertain	1	1	0.9488	0.8848	1.94E-2	2.41E-1
Modality Conflicted	0	0	0.9999	0.0411	4.66E1	1.50E-13

pred.=prediction. Throughout this section, class labels are defined as: 0 represents fake news and 1 represents real news. class0 evid., class1 evid.=evidence from evidence head before final classifier

Table 6 Belief and evidence values for Class 0 and Class 1 across different modalities and feature types

Type	**Text**				**Image**			
	Semantic		Pattern		Semantic		Pattern	
	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1
Belief	0.1906	0.1993	0.1936	0.2233	0.2023	0.2025	0.1855	0.2312
Evidence	0.6247	0.6534	0.6638	0.7656	0.6800	0.6807	0.6359	0.7925

exceeding 0.64 in all branches of Table 6. The accompanying image clearly depicts a spider-like insect with seven visible legs, matching the textual assertion. This semantic congruence contributes to a decisive and interpretable prediction. It demonstrates that TrustFND's strength in handling factual content where multimodal signals are well-aligned.

4.5.2 Confidently incorrect prediction

The instance in Fig. 5b was incorrectly classified as fake with high confidence (0.9916) and very low uncertainty, making it a false positive as shown in Table 5. This reflects overconfident error. Text encoding gives stronger evidential pattern-level support (0.709+) than semantic (0.610). pattern-level evidential features in the image are stronger than semantic. Pattern-level cues (from both modalities) seem stronger than semantic in this sample as shown in the Tables 5 and 7.

4.5.3 Correct but uncertain prediction

This example as shown in Fig. 5c is correctly predicted as real, but the confidence is not very high (0.9488) and uncertainty is relatively high (0.8848), indicating ambiguity. Text Modality related Belief mass has a Moderate value. Evidence-level features has Moderate values (0.65–0.69) that suggests some cues supported class 1, but not strongly. Image Modality related Belief mass has features Similar moderate activation. Evidence features Moderate evidence (0.65–0.67), with pattern-based evidence being higher, showing reliance on visual patterns. Low evidence values especially class 1 evidence = 0.24 suggests the extraction modules did not extract highly convincing patterns. Despite this, the model still leaned towards class 1, probably due to a relatively higher contrast between class 1 evidence and class 0 evidence (refer Tables 5 and 8).

4.5.4 Modality conflicted prediction

The sample as shown in Fig. 5d exhibits a modality conflict scenario because the text semantic features indicate high belief in Class 0 (supported by strong evidence), while image features (especially pattern-level) indicate a stronger belief in Class 1, also backed by decent evidence. The text pattern even flips sides compared to text semantic, adding to internal conflict within the text modality itself. These four cases demonstrate TrustFND's ability to not only detect misinformation but also to flag ambiguous or conflicting inputs. The framework's interpretability allows for understanding modality influence and managing prediction uncertainty-critical for real-world deployment as shown by the figures in Tables 5 and 9.

4.6 Analysis of modality contribution and prediction behavior

The stacked bar chart in Fig. 6 reveals that text evidence consistently contributes more than image evidence to the model's predictions, regardless of whether the predicted class is fake or real. This indicates that the text modality is the dominant factor influencing the final decision. However, image evidence also provides a meaningful contribution, highlighting the value of incorporating visual information. While not as influential as text, image cues enhance the model's ability to capture subtle correlations and reinforce or challenge textual claims. This finding supports the effectiveness of using multi-modal learning, where both modalities work together. The model does not ignore visual signals but rather combines them in a way that complements the dominant textual reasoning. Overall, this suggests a balanced fusion mechanism, where text drives the prediction, and images support interpretability and robustness, especially in edge cases or ambiguous content.

The distribution of prediction cases across confidence-accuracy categories for image- and text-dominant inputs can be seen in Figure 6. Most predictions fall under Safe—Correct (90 image-dominant, 95 text-dominant), indicating strong reliability of the multi-modal model. However, overconfident-wrong cases (23 and 18) reveal instances where high confidence accompanies incorrect predictions, posing trustworthiness concerns, particularly for image-dominant scenarios. Risky-Correct cases (17 and 13) suggest occasional under-confidence in correct predictions, while Uncertain-Wrong errors (5 and 6) are minimal, showing effective uncertainty alignment. Overall, the model performs robustly, though mitigating overconfident errors remains a key area for improvement.

While the proposed model demonstrates strong performance, several limitations remain. Future work could explore variable modality dropout to improve robustness against missing or corrupted inputs. Extending the framework to handle nuanced misinformation types, such as distinguishing satire from propaganda, and incorporating temporal or user-centric context, may enhance generalization. Additionally, improving uncertainty calibration under domain shifts remains an open challenge, warranting further investigation.

4.7 Qualitative analysis of cross-modal attention and evidential distribution in TrustFND

We conduct a qualitative analysis using example from the Fakeddit benchmark. This analysis focuses on how the model attends to and integrates multimodal cues when making predictions, as well as how uncertainty is distributed across modalities.

Text: "The shape of this goats' cheese looks like mini marshmallows."

Table 7 Belief and Evidence values for Class 0 and Class 1 across different modalities and feature types

Type	Text				Image			
	Semantic		Pattern		Semantic		Pattern	
	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1
Belief	0.3021	0.3032	0.2001	0.1871	0.2023	0.2019	0.1968	0.2188
Evidence	0.6530	0.6105	0.6619	0.7097	0.6791	0.6777	0.6733	0.7487

Table 8 Belief and Evidence values for Class 0 and Class 1 across different modalities and feature types

Type	Text				Image			
	Semantic		Pattern		Semantic		Pattern	
	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1
Belief	0.1895	0.2008	0.1804	0.2119	0.2024	0.2020	0.1835	0.2301
Evidence	0.6215	0.6585	0.5939	0.6975	0.6794	0.6782	0.6259	0.7846

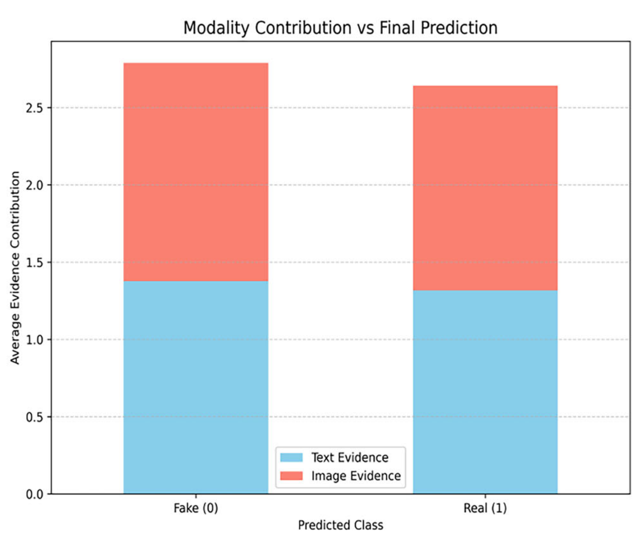
Table 9 Belief and Evidence values for Class 0 and Class 1 across different modalities and feature types

Type	Text				Image			
	Semantic		Pattern		Semantic		Pattern	
	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1	Class 0	Class 1
Belief	0.3209	0.3056	0.2283	0.1813	0.2881	0.3318	0.1826	0.2330
Evidence	0.7733	0.6140	0.6818	0.7827	0.6792	0.6780	0.6248	0.7974

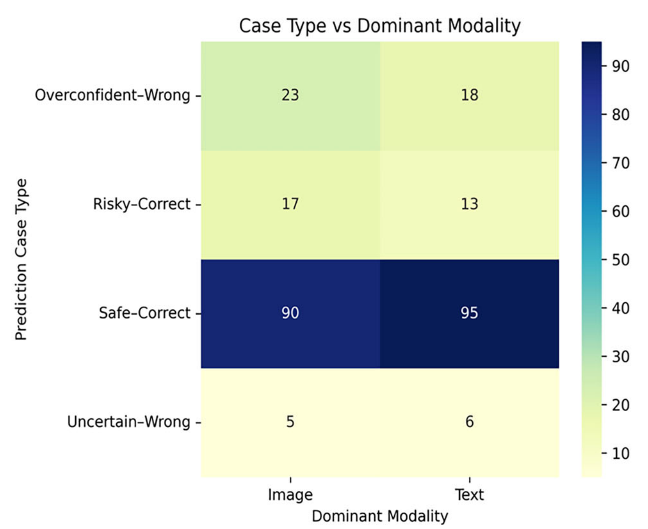
True Label: 1 **Predicted Label**:1 **Confidence**:0.9046 **Uncertainty**:0.1477

4.7.1 Text-to-image attention visualization

The text-to-image attention heatmap (refer Figure 7) shows how each text token focuses on specific image regions. Generic tokens like "the" and "shape" display broad attention patterns across the product package, while domain-specific tokens "goats" and "cheese" concentrate attention directly on the product label where these words appear. Critically, the token "looks" heightens attention toward the transparent packaging displaying the actual cheese, correctly linking the comparative verb to the visual object. This precise alignment between textual claims and corresponding visual features val-



(a) Average Modality Contribution Across Prediction Outcomes



(b) Case Type Distribution with Respect to Dominant Modality

Fig. 6 Modality contribution analysis. **a** Average evidence contribution from image and text modalities across predicted classes Fake (0) and Real (1). **b** Heatmap showing the distribution of prediction types (Safe-Correct, Risky-Correct, Uncertain-Wrong, Overconfident-Wrong) by dominant modality (Image vs. Text)

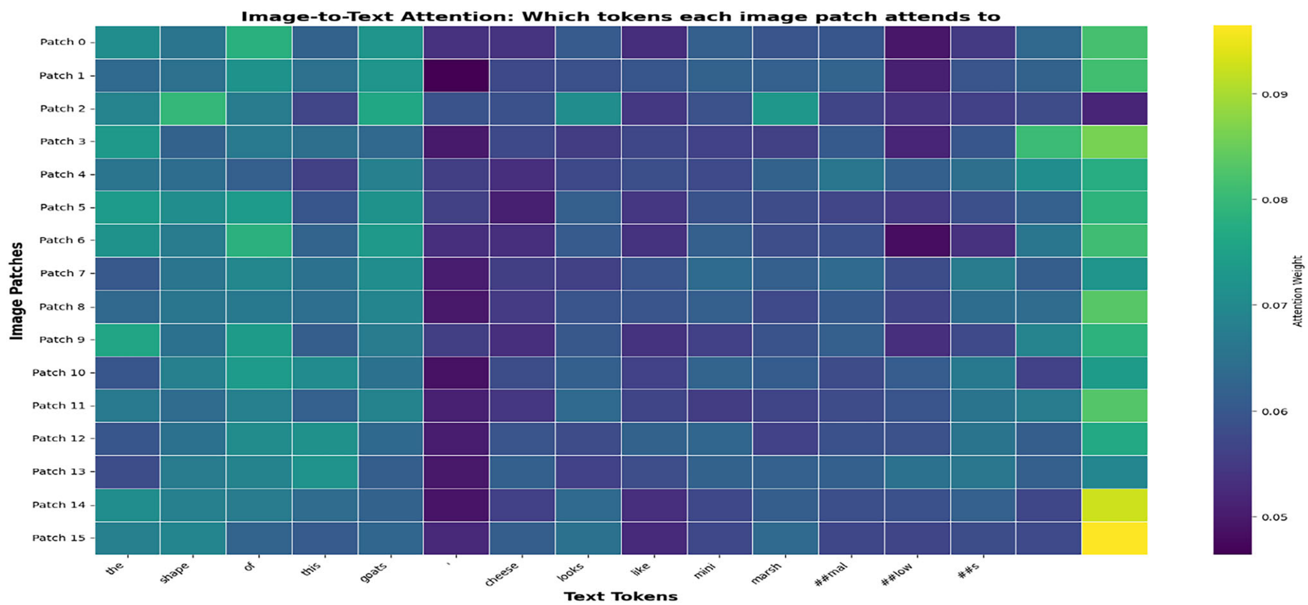


Fig. 7 Image-to-text attention heatmap

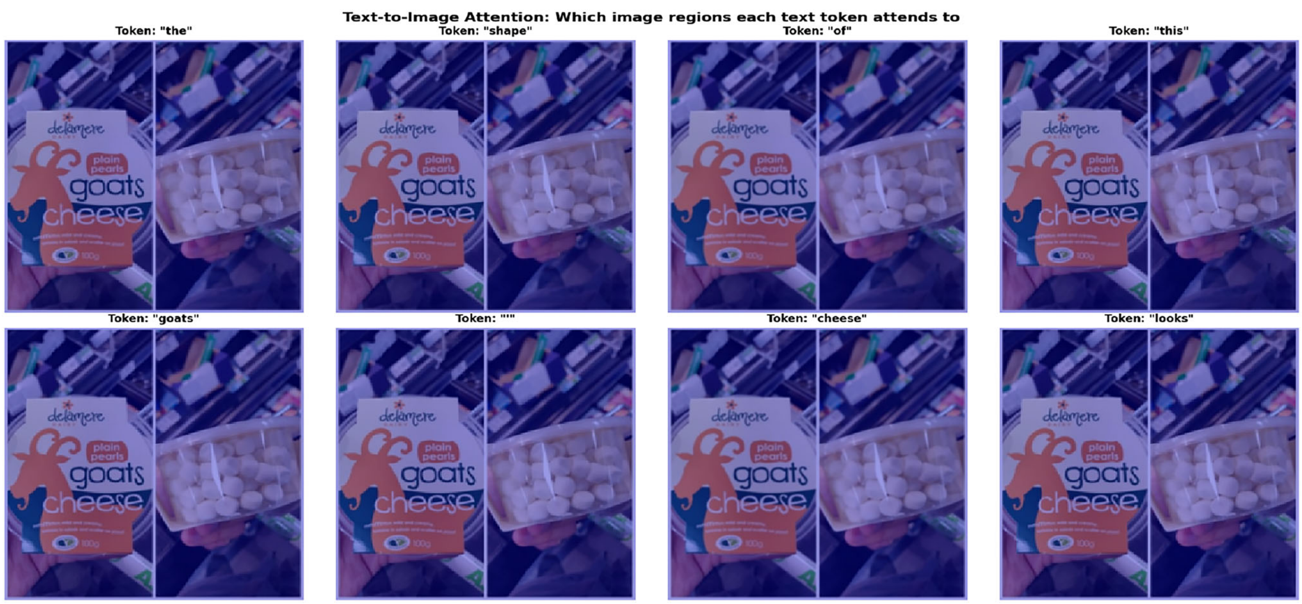


Fig. 8 Text-to-image attention visualization.

idates semantic consistency–a hallmark of authentic content absent in fake news where text-image relationships are fabricated or contradictory.

4.7.2 Image-to-text attention heatmap

Image patches 0-13 exhibit uniform attention (teal/blue, 0.06-0.08) across descriptive tokens like "shape," "goats," and "cheese," indicating general product information processing(refer Figure 8). Patch 14 shows dramatically elevated attention (bright yellow,≈0.10) toward final tokens " ml" and " ow," suggesting this patch captures specific visual evidence supporting the marshmallow comparison. This hierarchical attention gradient–from global context to fine-grained detail–reflects coherent visual reasoning where evidence systematically supports textual descriptions, unlike fake news where attention patterns exhibit confusion or misalignment.

4.7.3 Correlation matrices (G_I and G_T)

The inter-modal correlation matrices (refer Figure 9) reveal structured cross-modal dependencies. The Image Cor-

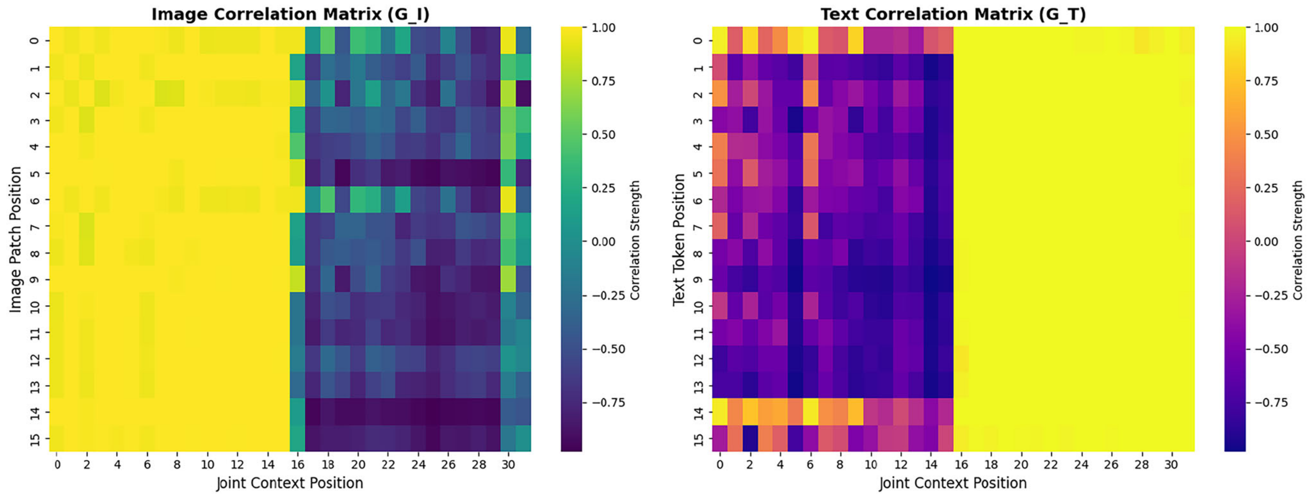


Fig. 9 Inter-modal fusion architecture

relation Matrix (G_I) shows strong correlations (yellow bands) at positions 0-15 and 16-31, with darker purple regions indicating semantic boundaries between modalities. The Text Correlation Matrix (G_T) exhibits complementary patterns with strong boundary correlations and consistent central-region patterns (positions 8-24). These organized correlation structures indicate systematic cross-modal alignment characteristic of authentic content, contrasting with the chaotic patterns expected in fake news where text and images are semantically misaligned.

4.7.4 Evidence distribution analysis

Both text evidence (semantic: 0.75, pattern: 0.77) and image evidence (semantic: 0.67, pattern: 0.70) strongly favor "real", producing overwhelming fused Dirichlet support ($\alpha \approx$12.4 vs. 0.9). The uncertainty progression reveals effective fusion: individual uncertainties (0.57-0.59) drop substantially after intra-modal DST combination (0.37-0.38) and further reduce after cross-modal fusion (0.15). This systematic uncertainty reduction indicates mutually reinforcing evidence from both modalities–characteristic of authentic content where multiple sources converge, contrasting with fake news that exhibits persistent high uncertainty due to semantic inconsistencies(refer Figure 10).

4.7.5 Model prediction justification

The TrustFND model correctly identified this sample as real (90.46% confidence, 14.77% uncertainty) through convergent multimodal evidence. Attention visualizations demonstrate precise semantic grounding: textual tokens directly attend to corresponding visual features (e.g., "cheese" $\rightarrow$ product label, "looks" $\rightarrow$ visible product). Correlation matrices reveal structured cross-modal alignment rather than chaotic patterns. Most critically, both modalities independently generate strong "real" evidence (0.70-0.77), which DST fusion combines into overwhelming support ($\alpha \approx 12.4$) with minimal conflict. The uncertainty reduction from 0.57 to 0.15 across fusion stages indicates mutually reinforcing cross-modal consistency–the signature of authentic content where descriptions match visuals. This systematic alignment across attention patterns, correlation structures, and evidence distributions validates the model's interpretable, trustworthy decision-making process for distinguishing genuine news from fabricated content with misleading text-image pairings.

5 Conclusion, limitations, and future Work

This study presents TrustFND, a robust multimodal framework for fake news detection that addresses key challenges in evidential reasoning and uncertainty quantification. By integrating textual and visual modalities with hierarchical evidence modeling, gated intra-modal DST fusion, and joint correlation refinement, TrustFND enables adaptive and interpretable predictions. The framework quantifies aleatoric and epistemic uncertainties, enhancing trust and reliability for high-stakes applications.

Experimental validation on the Fakeddit and Twitter datasets demonstrates that TrustFND achieves strong and balanced classification performance, with notable improvements in uncertainty calibration and interpretability compared to state-of-the-art baselines. The qualitative analysis further illustrates TrustFND's ability to distinguish correct predictions confidently, identify uncertain cases for review, and handle modality conflicts–with per-sample belief masses and uncertainty scores aiding transparent decision-making.

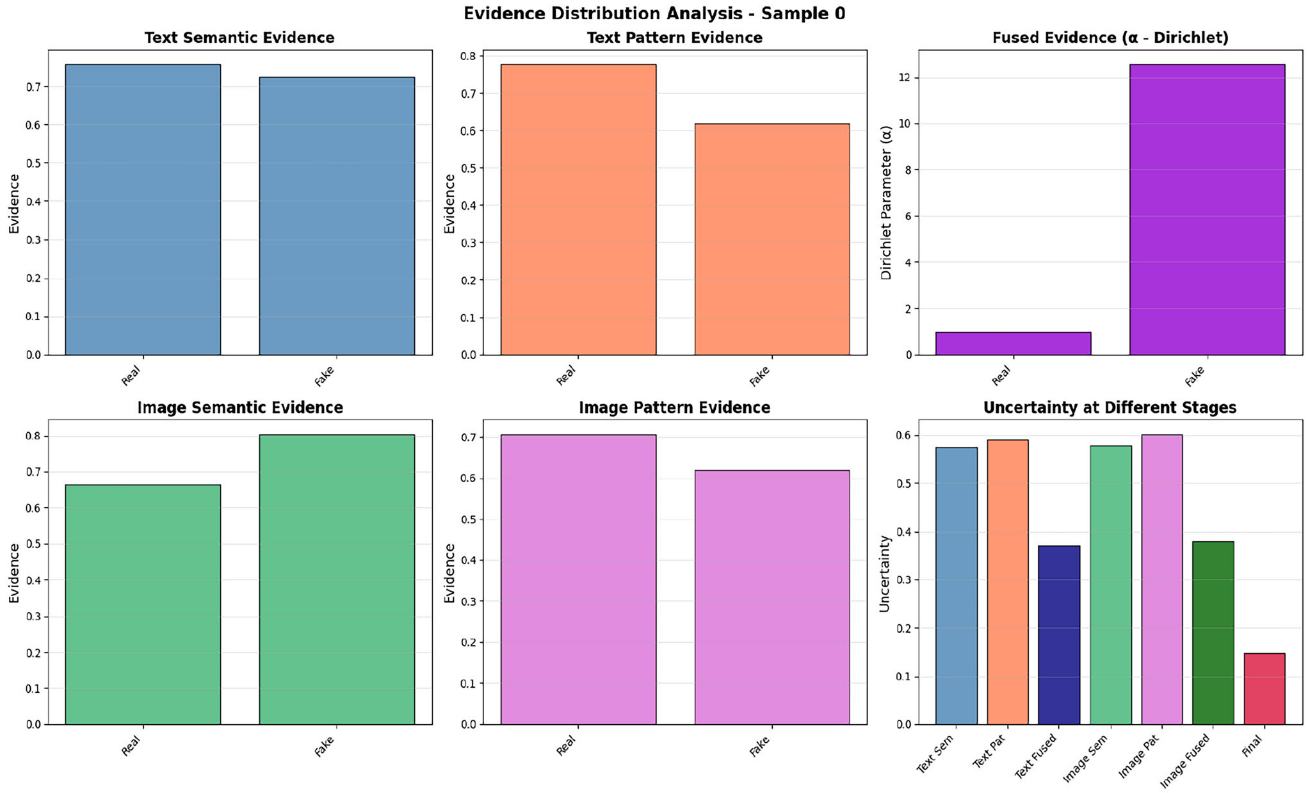


Fig. 10 Evidence distribution analysis

Limitations

Despite these advances, several limitations remain. The framework occasionally produces confidently incorrect predictions in cases with ambiguous or adversarial inputs, highlighting sensitivity to deceptive content and modality conflicts. Cross-modal disagreement can increase uncertainty or error, revealing that further refinement is needed to resolve contradictory evidence fully. In addition, current evaluations are limited to image-text scenarios, and the generalizability to video, audio, or multi-lingual contexts is yet to be explored. Scalability and adaptation to highly dynamic social media environments also present ongoing challenges.

Future work

Future research will address these limitations by enhancing the handling of modality conflicts and ambiguous cases through more nuanced evidential fusion and reasoning. Plans include extending TrustFND to incorporate additional modalities such as video and audio, investigating robustness against adversarial attacks, and improving adaptation to emerging misinformation formats. Further directions are expanding to multi-lingual, cross-domain datasets and collaborating with human moderators to integrate feedback for better transparency. Advancing uncertainty calibration and interpretability remains central, with ongoing work to ensure trustworthy deployment in real-world settings.

Author Contributions P.M. — Writing — Review & Editing, Visualization. R.T.– Conceptualization, Supervision, Methodology. J.P.T.K.J. – Data Curation, Software, Formal Analysis, Validation, Investigation. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Code availability The source code used in this study will be made publicly available upon acceptance of the manuscript.

Competing interests The authors declare no competing interests.

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